4MA1_1H_ch2_equations_formulae_indentities

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1. 4MA1/1H_Winter_2024_Q15 — 1 point

Write $3x(2x-1)(5x+4)$ in the form ax^3+bx^2+cx where a, b and c are integers.

3 marks

2. 4MA1/1H_Winter_2024_Q14 — 1 point

B is inversely proportional to the square of d

$B = 0.25$ when $d = 12$

Find a formula for B in terms of d

3 marks

3. 4MA1/1H_Winter_2024_Q8 1 point

(a) (i) Factorise $x^2 + 5x - 24$

....
(2)

(ii) Hence, solve $x^2 + 5x - 24 = 0$

....
(1)

(b) Solve the inequality $3y + 5 > 7y - 10$

Show clear algebraic working.

....
(3)

4. 4MA1/1H_Winter_2024_Q3 1 point

(a) Simplify $\left(p^3\right)^5$

....
(1)

(b) Expand and simplify $2n(4n + 3) + n(n - 4)$

....
(2)

(c) Solve $\frac{2x + 5}{3} = 4 - x$

Show clear algebraic working.

$x =$ ....
(3)

5. 4MA1/1HR_Summer_2024_Q20 1 point

(a) Express $2x^2 - 11x + 9$ in the form $a(x-b)^2 - c$ where a, b and c are numbers to be found.

(3)

The curve **C** has equation $y = 2(x-3)^2 - 11(x-3) + 9$

The point P is the minimum point on **C**

(b) Find the coordinates of P

(2)

6. 4MA1/1HR_Summer_2024_Q17 1 point

(a) $\left(\sqrt[4]{k^{12}}\right)^5 = k^n$

Find the value of n

(1)

(b) Express $\dfrac{7}{2-\sqrt{3}}$ in the form $\sqrt{c} + d$ where c and d are integers.

Show your working clearly.

(3)

7. 4MA1/1HR_Summer_2024_Q14 1 point

(a) Expand and simplify $(3x+1)(2-x)(4+x)$

(3)

(b) Simplify fully $\left(\dfrac{a^3b}{a^9b^5}\right)^{-\frac{1}{2}}$

(3)

8. 4MA1/1HR_Summer_2024_Q12 1 point

Solve the simultaneous equations

$$4x + 3y = 9.6$$
$$6x + 5y = 16.8$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(4)

9. 4MA1/1HR_Summer_2024_Q7 1 point

(a) Simplify $g^9 \div g^2$

(1)

(b) Expand $5k^2(k^3 + 4)$

(2)

(c) (i) Factorise $x^2 - 2x - 63$

(2)

(ii) Hence, solve $x^2 - 2x - 63 = 0$

(1)

(d) Solve the inequality $7 - 2y < 3y - 12$

(3)

10. 4MA1/1H_Summer_2024_Q25 1 point

$f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where a, b and c are constants.

$f(x) =$ ....

(4)

11. 4MA1/1H_Summer_2024_Q12 1 point

(a) Factorise fully $6y^2 - 5y - 4$

(2)

(b) Express $\dfrac{2x+1}{4x} + \dfrac{7-5x}{3x}$ as a single fraction in its simplest form.

(3)

12. 4MA1/1H_Summer_2024_Q10 1 point

(a) Simplify $(2p)^0$ where $p > 0$

(1)

$y^9 \times y^{-3} = y^n$

(b) Find the value of n

(1)

(c) Simplify fully $(5a^4c^2)^3$

(2)

13. 4MA1/1H_Summer_2024_Q8 1 point

A cylinder is placed on a table.

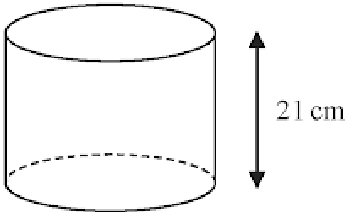


Diagram **NOT** accurately drawn

The volume of the cylinder is $1575\,\text{cm}^3$
The force exerted by the cylinder on the table is 84 newtons.

$$\boxed{\text{pressure} = \frac{\text{force}}{\text{area}}}$$

Work out the pressure on the table due to the cylinder.

.... newtons/cm^2

(3)

14. 4MA1/1H_Summer_2024_Q6 1 point

(a) Expand and simplify $(m + 5)(m - 8)$

(2)

(b) Solve $3n - 4 = \dfrac{5n + 6}{3}$

Show clear algebraic working.

$n =$ ....

(3)

15. 4MA1/1HR_Winter_2023_Q19 1 point

P is inversely proportional to y^2
When $y = 4$, $P = a$

(a) Find a formula for P in terms of y and a

....
(3)

Given also that y is directly proportional to $\sqrt{x}$
and when $x = a$, $P = 4a$

(b) find a formula for P in terms of x and a

....
(3)

16. 4MA1/1HR_Winter_2023_Q18

1 point

Solve $\sqrt{3}(x - 2\sqrt{3}) = x + 2\sqrt{3}$

Give your answer in the form $a + b\sqrt{3}$ where a and b are integers.
Show your working clearly.

$x =$ ....

4 marks

17. 4MA1/1HR_Winter_2023_Q13 1 point

Use algebra to show that $0.3\dot{8}\dot{1} = \frac{21}{55}$

2 marks

18. 4MA1/1HR_Winter_2023_Q12 1 point

Solve $2^{-4x} = 32$

$x =$ ....

2 marks

19. 4MA1/1HR_Winter_2023_Q8 1 point

(a) Factorise fully $18c^3d^2 - 21c^2$

....

(2)

(b) (i) Factorise $y^2 - 3y - 18$

....

(2)

(ii) Hence, solve $y^2 - 3y - 18 = 0$

....

(1)

20. 4MA1/1HR_Winter_2023_Q6 1 point

Solve the simultaneous equations

$$\begin{aligned} x + 2y &= 15 \\ 4x - 6y &= 4 \end{aligned}$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

3 marks

21. 4MA1/1H_Winter_2023_Q17 1 point

y is inversely proportional to $\sqrt{x}$

$y = c^4$ when $x = c^2$ where c is a positive constant.

Find a formula for y in terms of x and c
Give your answer in its simplest form.

....

3 marks

22. 4MA1/1H_Winter_2023_Q13 1 point

Expand and simplify $3x(2x-5)^2$

Show clear algebraic working.

....

3 marks

23. 4MA1/1H_Winter_2023_Q12 1 point

$3^{\frac{1}{2}} \times 3^{\frac{2}{5}} = 3^m$

(a) Work out the value of m

$m =$ ....

(1)

$5^{-10} \div 5^{-4} = 5^n$

(b) Work out the value of n

$n =$ ....

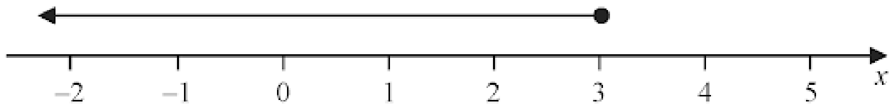
(1)

(a) Factorise $y^2 - 2y - 48$

....

(2)

(b) Write down the inequality shown on the number line



....

(1)

(c) Solve the inequality $7w + 6 > 12w + 14$

....

(3)

25. 4MA1/1H_Winter_2023_Q2 1 point

Solve $3(2 - 4x) = 5 - 8x$
Show clear algebraic working.

$x =$ ....

3 marks

26. 4MA1/1HR_Summer_2023_Q24 1 point

Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for x in terms of n
State any restrictions on n

....

5 marks

27. 4MA1/1HR_Summer_2023_Q23 1 point

Simplify $\left(x^2 - 4\right) \div \left(\frac{4x^2 - 7x - 2}{x}\right) - 2x$

Give your answer in the form $\frac{ax^2}{bx + c}$ where a, b and c are integers.

....

4 marks

28. 4MA1/1HR_Summer_2023_Q17 1 point

F is inversely proportional to the square of r

$F = 36$ when $r = 4$

(a) Find a formula for F in terms of r

....

(3)

(b) Work out the value of F when $r = 48$

....

(1)

29. 4MA1/1HR_Summer_2023_Q16 1 point

Use algebra to show that $0.1\dot{7}\dot{6} = \frac{35}{198}$

2 marks

30. 4MA1/1HR_Summer_2023_Q7

1 point

(a) Expand $4x(x-5)$

....

(1)

(b) Factorise $y^2 - 9y + 20$

....

(2)

31. 4MA1/1HR_Summer_2023_Q6 1 point

(a) Simplify $m^{10} \div m^3$

....
(1)

$k^n \times k^4 = k^{12}$

(b) Write down the value of n

$n =$ ....
(1)

(c) Simplify $(3x^6y^8)^2$

....
(2)

32. 4MA1/1H_Summer_2023_Q14

1 point

Factorise fully $50g^2 - 18$

....

3 marks

33. 4MA1/1H_Summer_2023_Q9 1 point

Solve the simultaneous equations

$$2x + 9y = 14.5$$
$$7x + 3y = 8$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

3 marks

34. 4MA1/1H_Summer_2023_Q6 1 point

(a) Simplify $(2c^4d^7)^3$

....
(2)

(b) Find the value of $5y^0$ where $y > 0$

....
(1)

(c) Factorise fully $16a^2b^3 + 20a^3b$

....
(2)

(d) (i) Factorise $x^2 + 9x - 22$

....
(2)

(ii) Hence solve $x^2 + 9x - 22 = 0$

....
(1)

35. 4MA1/1HR_Winter_2022_Q19 1 point

Given that $\left(\sqrt[3]{\frac{1}{x}}\right)^4 = x^m$

(a) find the value of m

$m =$ ....

(1)

Given that a, b and c are integers,

(b) express $3x^2 + 12x + 19$ in the form $a(x + b)^2 + c$

....

(2)

36. 4MA1/1HR_Winter_2022_Q15 1 point

Using algebra, prove that, given any 3 consecutive whole numbers, the sum of the square of the smallest number and the square of the largest number is always 2 more than twice the square of the middle number.

3 marks

37. 4MA1/1HR_Winter_2022_Q12 1 point

(a) Expand and simplify $n(n-4)(3n+5)$

....

(2)

(b) Express

$$\frac{3}{x}+\frac{x+2}{2x}+\frac{1}{4}$$

as a single fraction in its simplest form.

....

(3)

38. 4MA1/1HR_Winter_2022_Q9 1 point

(a) Simplify $x^4 \times x^5$

....
(1)

(b) Simplify $(4y^2)^3$

....
(2)

(c) Factorise $n^2 - 7n + 12$

....
(2)

39. 4MA1/1H_Winter_2022_Q20 1 point

(a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a, b and c are integers.

....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$
The point A is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of A

(.................. , ...)
(1)

40. 4MA1/1H_Winter_2022_Q19 1 point

Solve the simultaneous equations

$$3x^2 + y^2 - xy = 5$$
$$y = 2x - 3$$

Show clear algebraic working.

....

5 marks

41. 4MA1/1H_Winter_2022_Q15 1 point

A is inversely proportional to C^2

$A = 40$ when $C = 1.5$

Calculate the value of C when $A = 1000$

$C =$ ....

3 marks

42. 4MA1/1H_Winter_2022_Q13 1 point

(a) Expand and simplify $5x(x + 2)(3x - 4)$

....
(3)

(b) Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$

....
(3)

43. 4MA1/1H_Winter_2022_Q6 1 point

(a) Solve the inequality $5x - 7 \leqslant 2$

....

(2)

(b) (i) Factorise $y^2 - 2y - 35$

....

(2)

(ii) Hence, solve $y^2 - 2y - 35 = 0$

....

(1)

44. 4MA1/1H_Winter_2022_Q1 1 point

(a) Simplify $a^7 \times a^4$

....

(1)

(b) Simplify $w^{15} \div w^3$

....

(1)

(c) Simplify $(8x^5y^3)^2$

....

(2)

(d) Make t the subject of $c = t^3 - 8v$

....

(2)

45. 4MA1/1HR_Summer_2022_Q23 1 point

(a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a, b and c are integers.

....

(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point M is the minimum point on **C**

(b) Find the coordinates of M

(.................... , ...)

(2)

46. 4MA1/1HR_Summer_2022_Q16 1 point

(a) Expand and simplify $(3x-1)(x+2)(3x+1)$

....

(3)

(b) Simplify fully $\left(\dfrac{2x^5}{8xy^2}\right)^{-2}$

....

(3)

47. 4MA1/1HR_Summer_2022_Q14 1 point

Solve the simultaneous equations

$$3x - 5y = 25$$
$$4x + 3y = 14$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

4 marks

48. 4MA1/1HR_Summer_2022_Q9 1 point

Solve the inequality $3 - 4x \leqslant 11$

....

2 marks

49. 4MA1/1HR_Summer_2022_Q5 1 point

(a) Expand $3c^3(c + 4)$

....
(2)

(b) (i) Factorise $x^2 + 8x - 9$

....
(2)

(ii) Hence, solve $x^2 + 8x - 9 = 0$

....
(1)

50. 4MA1/1H_Summer_2022_Q24 1 point

Express each of a, b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$a =$ ....

$b =$ ....

$c =$ ....

4 marks